

qPCRtools: a cross-platform C++ desktop application for qPCR data analysis with validated equivalence to its R implementation

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Abstract

Quantitative real-time PCR (qPCR) is a cornerstone technique for measuring gene expression, yet rigorous analysis—primer-amplification-efficiency estimation, choice of quantification method, statistical testing, and publication-quality visualization—remains non-trivial. Existing tools either require programming, notably in R (including our own *qPCRtools* package), or are commercial software that is costly, closed-source, and platform-restricted, or are web services that are often discontinued and raise data-privacy concerns. Here we present *qPCRtools*, a cross-platform desktop application written in C++ with Qt6 that delivers a complete qPCR analysis pipeline through a graphical interface and ships as a single executable with no runtime dependencies. The application implements standard-curve construction and amplification-efficiency estimation, the ΔC_t and $\Delta\Delta C_t$ methods, and an efficiency-based (RqPCR) method, together with extended statistical testing (independent, paired, and Welch t -tests, Wilcoxon tests, one-way ANOVA with Tukey’s HSD, Shapiro–Wilk normality testing, and Cohen’s d) and interactive, publication-quality visualization; the standard-curve slope–intercept quantification and reverse-transcription volume calculation of the R version are not yet ported. We validated the C++ implementation against the original R package on multiple real datasets: across all shared methods the results agree to numerical precision, and the validation is fully reproducible from open scripts. *qPCRtools* is freely available under the MIT license for Windows and macOS, with source code compilable on Linux.

1 Introduction

Quantitative real-time PCR (qPCR) is among the most widely used techniques in molecular biology for quantifying nucleic acids and measuring gene expression [4]. A sound qPCR analysis involves several interdependent steps: estimating primer amplification efficiency, choosing an appropriate quantification strategy, performing statistical testing, and producing publication-quality figures [3]. When the amplification efficiencies of target and reference genes are comparable, the $2^{-\Delta\Delta C_T}$ method [6] is convenient; when they differ, an efficiency-corrected or standard-curve-based approach is required instead [8]. These analytical choices, combined with the need for rigorous statistics and clear visualization, make qPCR data analysis demanding for many laboratory researchers.

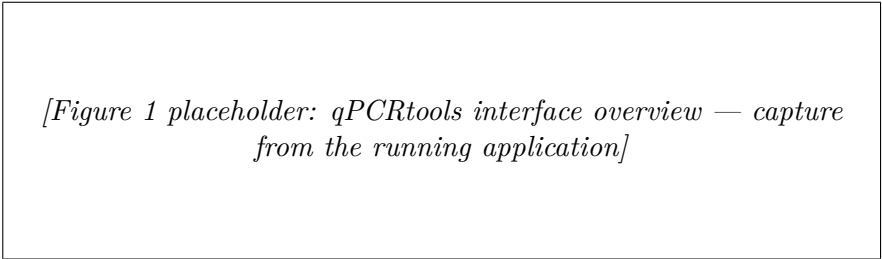
A range of tools address qPCR analysis [7], yet each carries trade-offs. R packages—including our own *qPCRtools* [5] and *pcr* [1]—implement sound algorithms but demand R programming skill and a configured software environment, which hampers reproducibility across collaborators. Commercial packages such as QuantStudio, CFX Maestro, REST, and qbase+ offer graphical interfaces but are costly, closed-source, and frequently restricted to a single operating system or instrument. Free web-based tools exist but are often discontinued and require uploading experimental data to remote servers, raising reproducibility and privacy concerns. The gap, therefore, is for a tool that combines *validated algorithmic equivalence* to a trusted implementation with a *native, cross-platform, zero-dependency* delivery, together with statistical options broader than those of the original R package.

We introduce *qPCRtools*, a C++/Qt6 desktop application that reimplements the analytical core of our R package behind a point-and-click interface distributed as a single executable. This Application Note makes four contributions: (i) a C++/Qt architecture in which a tested core algorithm library is decoupled from a web-based user interface; (ii) a validation demonstrating numerical equivalence to the R package across all shared quantification methods; (iii) statistical and functional extensions beyond the original R version, including non-parametric tests, normality testing, effect sizes, and reference-gene selection; and (iv) a fully reproducible validation package.

2 Implementation

2.1 Architecture

qPCRtools is written in C++17 and built on Qt6 [10], which provides cross-platform abstractions and the native application shell. The user interface is rendered with web technologies—Bootstrap 5.3 for layout and Apache ECharts [2] for interactive charts—hosted inside a `QWebView`, and exchanges data with the native backend through Qt `WebChannel` via a registered bridge object. This separation lets the presentation layer iterate rapidly as HTML, CSS, and JavaScript while all numerical computation stays in compiled native code. The analytical core is isolated as a standalone static library, `qpcr_core`, which depends only on Qt Core and Eigen and is independent of the GUI; it is exercised directly by the unit-test suite and is also linked by the command-line equivalence driver described in Results. Statistical distribution functions use the GNU Scientific Library (GSL) for accuracy. The resulting interface is shown in Figure 1.



[Figure 1 placeholder: qPCRtools interface overview — capture from the running application]

Figure 1: The qPCRtools desktop interface. A single cross-platform window hosts the web-based UI for data import, method and statistics configuration, and interactive results; no programming or runtime dependencies are required.

2.2 Quantification algorithms

The core reimplements the analytical methods of the R package. From a dilution series, amplification efficiency is obtained by ordinary least squares of C_q on $\log_{10}(\text{concentration})$, yielding slope s , intercept, R^2 , and a regression p -value; efficiency is

$$E = d^{-1/s} - 1, \tag{1}$$

where d is the dilution factor. Relative expression is computed by the ΔC_t and $\Delta\Delta C_t$ methods [6],

$$\text{RE} = 2^{-[(C_{q,\text{target}} - C_{q,\text{ref}})_{\text{treated}} - (C_{q,\text{target}} - C_{q,\text{ref}})_{\text{control}}]}, \quad (2)$$

and by an efficiency-based (RqPCR) method [9] in which per-sample quantities are $E^{C_{q,\text{min}} - C_q}$ and are normalized by the geometric mean of one or more reference genes [8].

2.3 Statistical analysis

For each target gene the application compares groups with independent, paired, or Welch two-sample t -tests, the Wilcoxon rank-sum or signed-rank test, or one-way ANOVA followed by Tukey’s HSD post-hoc letter grouping. It additionally reports Cohen’s d effect sizes, Shapiro–Wilk normality testing, and IQR-based outlier removal, and can select reference genes automatically by the geNorm algorithm [11]. These options extend the original R package, which provided t -test and ANOVA/Tukey testing only.

2.4 Software engineering and scope

qPCRtools is managed with CMake and ships with a unit-test suite. Visualization follows a ggplot2-inspired aesthetic [12] with configurable color palettes and significance annotations; the interface is bilingual (Chinese/English) and accepts both CSV and Excel input. The application is distributed as a single self-contained executable for Windows and macOS, requiring no R, Python, or runtime installation, and the source compiles on Linux. It is released under the MIT license. Two R functions are not yet ported—the standard-curve slope–intercept quantification (`CalExpCurve`) and the reverse-transcription reagent volume calculation (`CalRTable`)—and are planned for a future release.

3 Results

3.1 Typical analysis workflow

To illustrate a typical analysis, we analyzed the bundled example dataset (`examples/cq.csv` and `examples/design.csv`), comprising the target gene *fos-glo-myc* normalized against the reference gene *Beta Actin* across three treatment groups (0, 0.5, 1) with three biological replicates each. Using the

$\Delta\Delta C_t$ method with a two-sample t -test against the control group (0), relative expression rose monotonically to 1.11 at dose 0.5 and 1.59 at dose 1, both significant at $p < 10^{-3}$ (Figure 2). Applied to a four-fold dilution series of four genes, the standard-curve module recovered slopes between -2.07 and -2.24 with $R^2 > 0.99$ for three of the four primers (efficiencies 0.85–0.95), illustrating amplification-efficiency estimation that, per MIQE guidance, determines which quantification method is appropriate. The complete workflow—import, configuration, analysis, and export—is performed through the graphical interface without writing code.

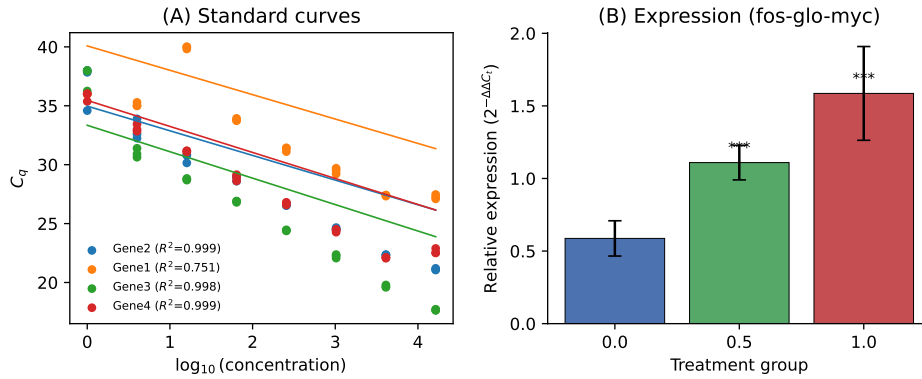


Figure 2: Analysis output on real data. (A) Standard curves for four genes over a four-fold dilution series, with fitted regressions and R^2 . (B) Relative expression of *fos-glo-myc* across treatment groups ($\Delta\Delta C_t$ method), with significance labels.

3.2 Equivalence to the R package

We validated the C++ implementation against the original R package [5] by running both on identical inputs for every shared method— $\Delta\Delta C_t$ (`CalExp2ddCt`), the standard curve (`CalCurve`), and the efficiency-based RqPCR method (`CalExpRqPCR`)—and comparing every resulting quantity pairwise (Table 1, Figure 3). Inputs were drawn from the repository’s example data and from a dilution series used in the original R package; both implementations read the same on-disk files, so any difference reflects the implementations alone.

For the two expression methods, $\Delta\Delta C_t$ and RqPCR, every compared quantity—expression values, standard deviations, and standard errors—matched between implementations to within floating-point precision (max-

Table 1: Equivalence of C++ and R results across all shared methods on identical inputs. Relative difference is $|R - C++|/|R|$.

Method	Compared quantities	Pairs	Max. rel. diff.
$\Delta\Delta C_t$	expression, SD	6	3.8×10^{-15}
Standard curve	slope, intercept, R^2 , efficiency	16	2.5×10^{-3}
RqPCR	expression, SE	16	2.1×10^{-15}

imum relative difference $< 4 \times 10^{-15}$). For the standard curve, slope, intercept, R^2 , and efficiency agreed to within 2.5×10^{-3} relative difference; the residual is attributable to the R package rounding its tabulated output to two to four decimal places, since the underlying C_q inputs are identical. Plotting every C++ result against its R counterpart, all points fall on the line $y = x$ (Figure 3). These results confirm that the C++ reimplementation preserves the analytical behavior of the R package across all shared methods. The validation is fully reproducible from the open scripts in `validation/` (see Availability).

4 Availability

The source code and pre-built binaries for Windows and macOS are available at https://github.com/lixiang117423/qPCRtools_cpp under the MIT license. The repository includes documentation, a unit-test suite, example datasets, and a fully reproducible equivalence-validation package (`validation/`) comprising the R script, a C++ command-line driver linked against `qpcr_core`, and a comparison script that regenerates all validation outputs and figures. The project builds with CMake (≥ 3.20), requires a C++17 compiler and Qt 6 (≥ 6.5) with the WebEngine and WebChannel modules, and optionally links Eigen3 and GSL.

Authors’ contributions

X.L. conceived and designed the software, implemented the C++ codebase, performed the equivalence validation, and wrote the manuscript.

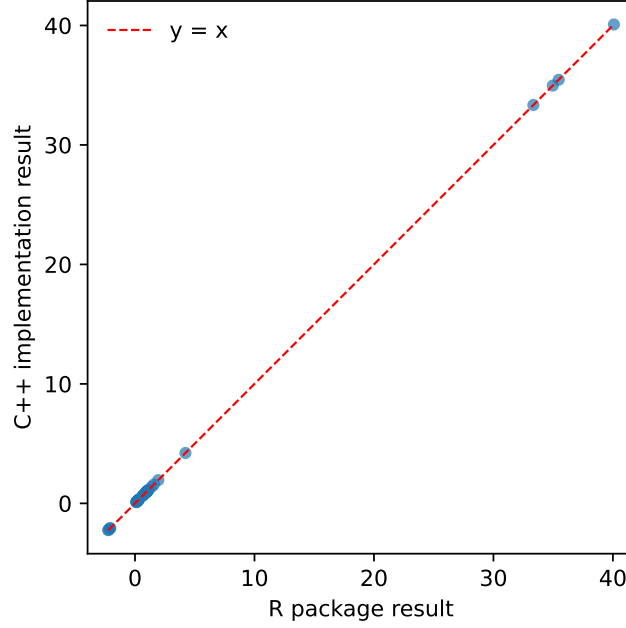


Figure 3: Numerical equivalence of the C++ implementation and the R package. Each C++ result is plotted against the corresponding R result across all shared methods ($\Delta\Delta C_t$, standard curve, RqPCR); all points fall on the $y = x$ line.

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Conflict of interest

The author declares no competing interests.

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